

Stephen Murphy

I help businesses deliver products to market through solid engineering design, ambitious goal-setting, effective communication between project teams, and positive work cultures.

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📍 San Diego, CA, USA

WORK EXPERIENCE

Sr. Staff Mechanical Engineer

Shield AI

05/2025 - Present

San Diego, CA

- Owned the conceptual development of the X-BAT wing fold mechanism which supports full 180° folding of both wings to achieve an unprecedented spotting factor of 0.33 on a 39' wingspan aircraft inclusive of mechanism, actuators, plumbing, lock pins, and doors all within mass budget.
- Designed the environmental cooling system (ECS) for the X-BAT Hopper prototype including heat exchangers, plumbing, manifolds, fluid storage, instrumentation, and air cooling.
- Personally developed CAD and drawings in Siemens NX and simulations in Simcenter.

Staff Mechanical Engineer

Shield AI

12/2023 - 04/2025

San Diego, CA

- Led the development of the V-BAT wing and fuselage modules from concept to production with novel designs for a cam-lock attachment mechanism, rotating pitot-static probe for easier storage, rib-free composite structure for reduced cost and assembly time, radio and antenna integrations, electrical harnessing, and PCBA integrations.

Mechanical Design Manager

SkySafe

01/2023 - 11/2023

San Diego, CA

- Optimized the mechanical and electrical design of 2 drone detection platforms for improved ease of manufacture, cost, and reliability
- Led and performed technical work including Finite Element Analysis (thermal, and structural), CFD, drawing reviews (ASME Y14.5-2009), compliance (RoHS & CE), and cross-functional communication

Senior Systems Engineer

Reverb Industrial

09/2022 - 01/2023

San Diego, CA

- Defined system architecture and completed the full mechanical and electrical design of the first Alpha prototype for a novel wire-based metal additive-manufacturing platform.

Design Manager, Advanced & New Product Development

SharkNinja

03/2021 - 09/2022

Boston, MA

- Owned the product design engineering of Shark FlexStyle, HyperAir, SpeedStyle, and SmoothStyle high volume consumer electronics products from PoC to mass production at volumes > 100k units/year
- Led and performed mechanical design work including CAD (CREO), electronics integration, injection molding, DFM, CFD, FEA, DFMEA, and tolerance analysis
- Promoted to US Personal Care Engineering Lead, where I led a team of 4 AD/NPD engineers and cross-functional representation

Software Engineering Manager / Software Engineer

Intel

09/2018 - 01/2020

St. Leon-Rot, Germany

- Programmed UIs in Java, JavaFX, and CSS
- Led a 9-person software team to the successful release of Intel Mission Control, a drone flight planning application for Windows

Senior Mechanical Engineer

Intel

10/2016 - 09/2018

St. Leon-Rot, Germany

- Developed a commercial UAS from scratch including a 2kg aircraft, batteries, GNSS base station, remote, and charger, resulting in my promotion to lead a team of 9 mechanical engineers

Mechanical Engineer

MAVinci

10/2015 - 10/2016

St. Leon-Rot, Germany

- Designed and built a custom 3-axis CNC and a 2kg fixed wing UAS

STRENGTHS

Industries

- Aerospace
- Consumer electronics
- Drones
- Telecom
- Metal additive manufacturing
- Batteries

CAD

- NX
- Creo
- SOLIDWORKS
- Fusion 360

Programming & Simulation

- MATLAB
- Abaqus FEA
- Simcenter 3D FEA
- COMSOL FEA & CFD
- Java
- Python
- Simulink
- STAR-CCM+ CFD

Mechanical Design

- Fluid systems
- Injection molding
- Sheet metal
- Tolerance analysis
- DFM/DFA
- Motors
- Acoustics
- Batteries
- Mechanisms
- Machining
- Composites
- DFMEA
- Airflow
- Thermal management
- Vibration
- Actuators

Leadership

- Engineering management
- Team building
- System architecture
- Technical ownership
- Cross-functional execution

EDUCATION

Master of Engineering in Mechanical Engineering

University of Victoria

2020

Project: Thermo-electrochemical Model of a Zinc-air Flow Battery

Bachelor of Engineering in

Mechanical Engineering with Co-op
University of Victoria 2015

Top 15% of class; honors thesis on high-altitude, long-endurance biplane design

LANGUAGES

English: Native speaker

German: Limited working proficiency

PERSONAL INTERESTS

- Cooking while listening to old soul
- Building and riding road bikes
- Ultralight hiking
- Tennis, Swimming, Yoga, Running